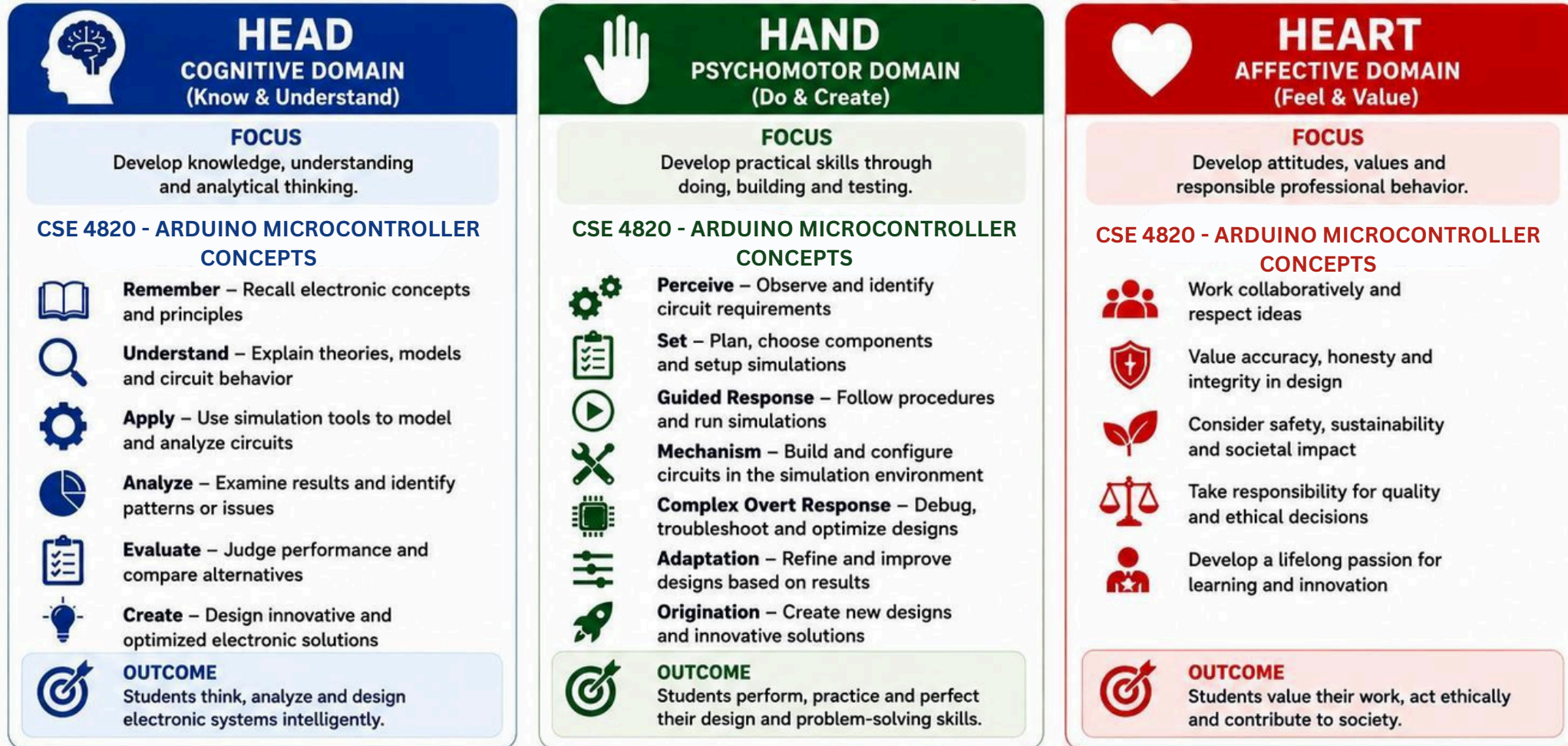


CSE 4820 - ARDUINO MICROCONTROLLER CONCEPTS

CONFORMS TO THE THREE DOMAINS OF LEARNING

One Subject – Three Domains – Complete Learning



THE POWER OF 3 DOMAINS



HEAD

Generates Ideas



HAND

Transforms Ideas into Reality



HEART

Adds Purpose and Responsibility



**COMPETENT
& RESPONSIBLE
ENGINEER**

KNOW IT. DO IT. VALUE IT. – THAT IS COMPLETE LEARNING.

3 DOMAINS OF LEARNING IN ACTION

CASE STUDY: TRAFFIC SIGNAL CONTROLLER

An automatic system that controls traffic safely at an intersection.



HEAD

**COGNITIVE DOMAIN
(THINK & UNDERSTAND)**

- 1 **Remember** – Recall concepts
- 2 **Understand** – Explain ideas
- 3 **Apply** – Use knowledge
- 4 **Analyze** – Examine and compare
- 5 **Evaluate** – Judge and assess
- 6 **Create** – Design new solutions

I KNOW



HAND

**PSYCHOMOTOR DOMAIN
(DO & CREATE)**

- 1 **Perceive** – Observe and identify
- 2 **Set** – Prepare and plan
- 3 **Guided Response** – Follow and practice
- 4 **Mechanism** – Perform with confidence
- 5 **Complex Overt Response** – Execute smoothly
- 6 **Adaptation** – Modify and adjust
- 7 **Origination** – Create and innovate

I CAN DO



HEART

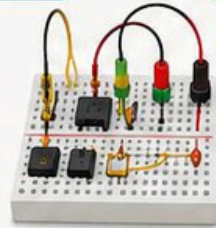
**AFFECTIVE DOMAIN
(FEEL & VALUE)**

- 1 **Receive** – Be aware and open
- 2 **Respond** – Participate actively
- 3 **Value** – Appreciate and show interest
- 4 **Organize** – Prioritize and align values
- 5 **Characterize** – Act consistently with values
- 6 **Internalize** – Make it part of life

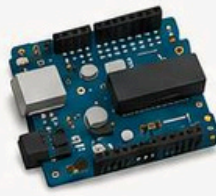
I CARE



**UNDERSTAND
THE CONCEPT**



BUILD & SIMULATE



IMPLEMENT



**SAFE ROADS
BETTER SOCIETY**



HEAD DOMAIN (COGNITIVE)

CASE STUDY: TRAFFIC SIGNAL CONTROLLER

The HEAD learns by THINKING – understanding, analyzing, evaluating and creating.



BLOOM'S TAXONOMY (COGNITIVE LEVELS)

- 1 REMEMBER**
Recall facts and basic concepts
- 2 UNDERSTAND**
Explain ideas and concepts
- 3 APPLY**
Use knowledge in real situations
- 4 ANALYZE**
Examine parts and relationships
- 5 EVALUATE**
Judge and compare alternatives
- 6 CREATE**
Design new solutions and improvements

FROM KNOWLEDGE
TO INNOVATION

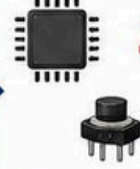
COGNITIVE ACTIVITIES IN THIS CASE

1. UNDERSTAND THE SYSTEM



- What is a traffic signal controller?
- Why is timing and safety important?

2. STUDY COMPONENTS & PRINCIPLES



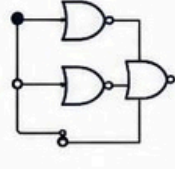
- LEDs, sensors, push button
- Microcontroller (Arduino)
- Digital logic & circuits

3. ANALYZE REQUIREMENTS



- Traffic flow patterns
- Timing sequences
- Pedestrian safety
- Constraints & conditions

4. DESIGN THE LOGIC



- State machine
- Timing diagram
- Flow chart / Algorithm

5. EVALUATE ALTERNATIVES



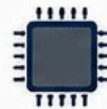
- Compare timing strategies
- Check efficiency
- Ensure safety & reliability

6. CREATE & IMPROVE SOLUTIONS

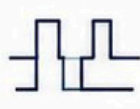


- Optimize timing
- Add features (sensors, emergency override)
- Smart traffic control

KEY CONCEPTS HANDLED BY THE HEAD



ELECTRONIC COMPONENTS
LEDs, Resistors, ICs, Sensors



DIGITAL LOGIC
Logic Gates, Flip-Flops, State Machine



ALGORITHMS & PROGRAMMING
Arduino Code, Control Logic



TIMING & SEQUENCING
Green/Yellow/Red Durations, Cycle Timing



SIMULATION
Test & Verify in TinkerCAD, Proteus, etc.



PERFORMANCE EVALUATION
Accuracy, Efficiency, Reliability



SAFETY & RELIABILITY
Fail-Safe Design, Robust Operation



OUTCOME:

The HEAD thinks, understands, analyzes and designs the traffic signal controller for safe, efficient and smart traffic management.





HAND DOMAIN (PSYCHOMOTOR)

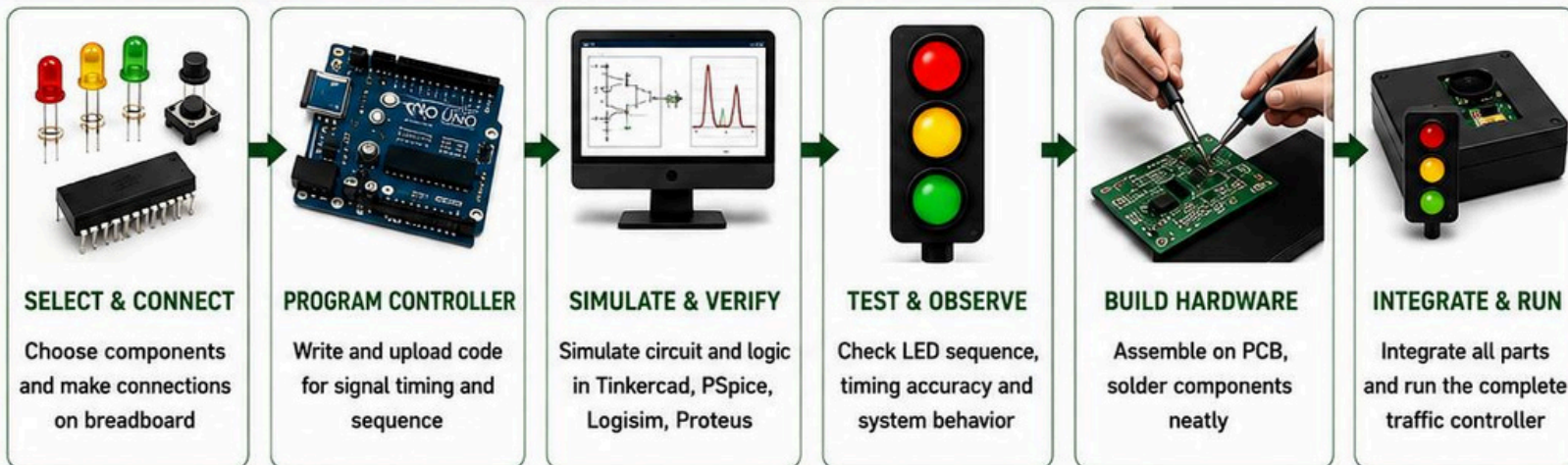
CASE STUDY: TRAFFIC SIGNAL CONTROLLER

The HAND learns by DOING – building, programming, testing and improving.

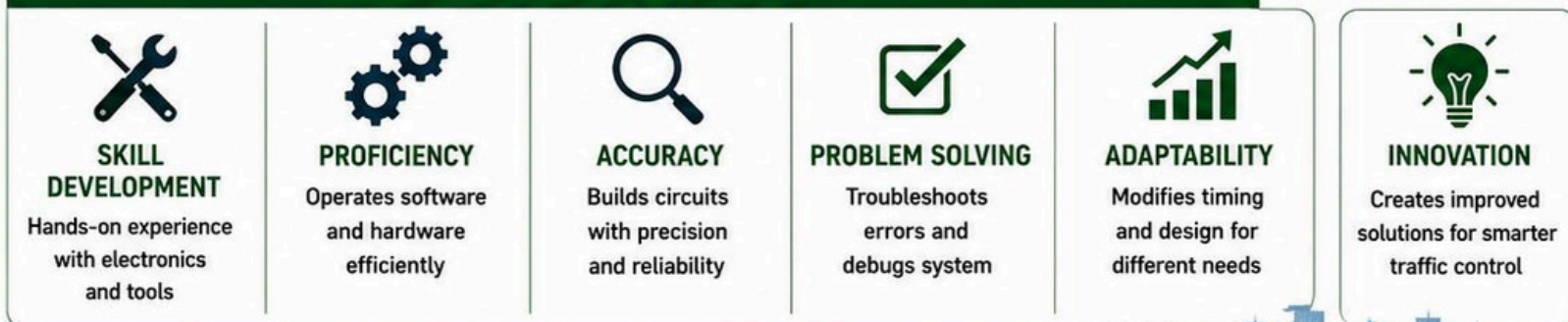
PSYCHOMOTOR TAXONOMY

- 1 PERCEPTION**
Recognize components and tools
- 2 SET**
Prepare workspace and plan steps
- 3 GUIDED RESPONSE**
Follow instructions and build with help
- 4 MECHANISM**
Perform tasks independently
- 5 COMPLEX OVERT RESPONSE**
Build and operate the traffic signal system
- 6 ADAPTATION**
Troubleshoot and modify the system
- 7 ORIGATION**
Design and implement improved solutions

PRACTICAL ASPECTS IN THIS CASE



WHAT THE HAND ACHIEVES



OUTCOME

THE HAND CAN BUILD, OPERATE, TROUBLESHOOT AND IMPROVE REAL-WORLD SYSTEMS WITH CONFIDENCE.





HEART DOMAIN (AFFECTIVE)

CASE STUDY: TRAFFIC SIGNAL CONTROLLER

The HEART learns by FEELING & VALUING – developing attitudes, values and responsibility.



KRATHWOHL'S AFFECTIVE TAXONOMY (5 LEVELS)

- 1 RECEIVING**
Be aware and willing to listen.
"I listen to the importance of traffic safety."
- 2 RESPONDING**
Actively participate and show interest.
"I take part in discussions and team work."
- 3 VALUING**
Value safety, quality and responsibility.
"I care about safe roads and public welfare."
- 4 ORGANIZATION**
Prioritize values and integrate into decisions.
"I prioritize safety, ethics and sustainability."
- 5 CHARACTERIZATION**
Live by values consistently; become part of character.
"I act responsibly and ethically in every project."

AFFECTIVE ASPECTS IN THIS CASE



SAFETY FIRST

Value human life.

- Design for pedestrian safety
- Reduce accidents and wait time



TEAMWORK

Work together with respect.

- Share ideas
- Support and learn from others



RESPONSIBILITY

Take ownership of your work.

- Meet deadlines
- Deliver reliable solutions



CARE FOR SOCIETY & ENVIRONMENT

Think beyond personal benefit.

- Energy efficient design
- Support smart and green cities



INTEGRITY & ETHICS

Be honest and follow the rules.

- No data falsification
- Respect IP and acknowledge others



LIFELONG LEARNING & IMPROVEMENT

Always seek to learn and improve.

- Accept feedback
- Upgrade skills and stay curious

WHAT THE HEART ACHIEVES



**SAFE ROADS
SAVE LIVES**



**BETTER TEAMWORK
BETTER SOLUTIONS**



**QUALITY WORK
YOU CAN TRUST**



**SUSTAINABLE
FUTURE**



**ETHICAL ENGINEERS
BETTER SOCIETY**



**GROWTH MINDSET
LIFELONG IMPACT**



OUTCOME:

The HEART builds values, shows care and responsibility, and works for the **SAFETY** and **WELL-BEING** of **SOCIETY**.



OBE INPUT DESIGN – ARDUINO MICROCONTROLLER CONCEPTS

Balanced 3-Domain Approach for a Practical Based Subject



PSYCHOMOTOR DOMAIN 35%
(Hand – Do & Create)



COGNITIVE DOMAIN 50%
(Head – Know & Understand)



AFFECTIVE DOMAIN 15%
(Heart – Feel & Value)

SUBJECT: SIMULATION BASED ELECTRONICS DESIGN

35% PSYCHOMOTOR INPUT

Students learn by **DOING, PRACTICING, BUILDING** and **SOLVING** using simulation tools.

	✓ Electronics Modeling in Simulation Tools	7%
	✓ Parameter Setup & Configuration	4%
	✓ Run Simulations & Observe Results	6%
	✓ Analyze Waveforms & Measurements	3%
	✓ Troubleshoot & Debug Circuits	5%
	✓ Optimize & Improve Performance	4%
	✓ Design New Circuits / Creative Solutions	6%



Outcome: Students can skillfully use simulation tools to design, test, analyze and optimize electronics systems independently.

50% COGNITIVE INPUT

Students learn by **THINKING, UNDERSTANDING, ANALYZING** and **EVALUATING** concepts.

	Concepts of Electronics Theory	12%
	Components & Their Behavior	8%
	Circuit Analysis Techniques	9%
	Simulation Principles & Methods	11%
	Result Interpretation & Evaluation	10%



Outcome: Students understand concepts, analyze electronics and make informed decisions in design and evaluation.

15% AFFECTIVE INPUT

Students learn by **VALUING, COLLABORATING** and **DEVELOPING** the right attitude.

	Teamwork & Collaboration	4%
	Responsibility & Discipline	2%
	Ethical Use of Tools & Data	3%
	Appreciation of Quality & Safety	1%
	Lifelong Learning & Curiosity	5%



Outcome: Students develop professional values, ethics, responsibility and a passion for continuous improvement.



OVERALL INPUT DISTRIBUTION

Psychomotor (Hand)	35%
Cognitive (Head)	50%
Affective (Heart)	15%
Total	100%

HOW IT WORKS



HAND (35%) : Builds practical skills through extensive simulation practice and problem solving.



HEAD (50%) : Strengthens understanding, analysis and decision making.



HEART (15%) : Shapes attitude, ethics and professional responsibility.



COMPETENT ENGINEER

★ The right blend of **HEAD, HAND** and **HEART** creates skilled, knowledgeable and responsible engineers. ★